

YUAN YUAN

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EDUCATION

The University of Sydney (USYD)

Doctor of Philosophy (Computer Science)

From 07 2020

Sydney, Australia

- **Research Areas:** Deep Learning, Computer Vision, Multimodal Medical Image Analysis
- **Supervisors:** Prof. Jinman Kim, Prof. Mohamed Khadra, Dr. Euijoon Ahn, Prof. Dagan Feng
- **Thesis:** Multimodal Medical Image Analysis for Cancer Diagnosis and Prognosis

Northwestern Polytechnical University (NPU)

Bachelor of Engineering (Information Engineering)

09 2015–06 2019

Xi'an, China

- **Thesis (External Collaboration):** Image Content Understanding Based on Geometric Structure (in collaboration with **Shanghai Jiao Tong University (SJTU)**)
- Outstanding Graduate (Top 5%) | GPA: 89.6/100

PROJECTS

USYD & Nepean Hospital

From 03 2022

Deep Learning for Prostate Cancer Detection and Diagnosis Using Multimodal MRI [Link] Sydney, Australia

- Proposed a Zonal-aware Self-supervised Mesh Network (Z-SSMNet) for clinically significant prostate cancer (csPCa) detection and diagnosis using bi-parametric MRI (bpMRI).
- The proposed network adaptively fused multiple 2D/2.5D/3D convolutional neural networks (CNNs) to capture a balanced representation of dense intra-plane information and sparse inter-plane information of bpMRI.
- Introduced a self-supervised learning technique to pre-train the network using large-scale unlabeled data.
- Proposed a network constraint technique to enhance csPCa detection by focusing on zonal anatomical regions.

Biomedical Data Analysis and Visualisation (BDAV) Lab, USYD

From 05 2023

Deep Learning for Focal Liver Lesions Diagnosis Using Multimodal MRI

Sydney, Australia

- Proposed a deep learning model for seven-class FLL classification using multiparametric MRI with 8 sequences.
- Proposed a Pyramid Lattice Network (PLN) to adaptively fuse 2D and 3D features for balanced learning of anisotropic mpMRI semantics.
- Proposed a multi-node adaptive attention module (MAAM) that dynamically selects and integrates features encoding different-level anisotropic information.
- Proposed a self-supervised pre-training technique to enable the model to learn the anisotropic features from the images using large-scale unlabeled data.

USYD & SJTU & Fudan University Shanghai Cancer Center

From 07 2020

Machine Learning for PET/CT-Based Analysis of Pancreatic Ductal Adenocarcinoma

Sydney & Shanghai

- Collaborated with clinicians to gather annotated PET/CT imaging data and clinical indicators for studies of pancreatic ductal adenocarcinoma (PDAC).
- Proposed a radiomics-based machine learning model for preoperative prediction of microvascular invasion and perineural invasion of PDAC.
- Proposed a deep learning-based multi-task learning model for joint tumor segmentation and recurrence prediction of PDAC.

Visual Media and Data Management Lab, SJTU

11 2018–08 2019

Image Content Understanding Based on Geometric Structure

Shanghai, China

- Proposed a parallel CNN architecture to recover albedo and shading for single image intrinsic decomposition in parallel.
- The image reconstruction loss function was shared across the parallel networks to maximize reconstruction consistency.
- Integrated gradient information into the CNN-based model, leveraging physics-based priors to better handle complex real-world scenarios.

Center of Intelligent Acoustics and Immersive Communications (CIAIC), NPU 12 2017–07 2018

Multimodal Emotion Recognition Using EEG and Speech Signals

Xi'an, China

- Studied speech signal processing and analytical methods, applying SVM, ELM, and GMM for discrete-speech emotion classification. Collected data for model training, validation, and testing.
- Studied EEG theory, recorded and processed EPR statistics, conducted EEG experiments, and analyzed data using Curry 8/EEGLAB, applying PCA for feature dimensionality reduction.
- Studied emotion recognition using EEG and speech signals, applied ANOVA for feature selection, and analyzed the impact of environmental deviation and sentiment on emotion classification.

WORK EXPERIENCES

The University of Sydney

Casual Academic (Tutor)

From 02 2025
Sydney, Australia

- Tutor the course ENGG1810 – Introduction to Engineering Computing.

From 12 2024

Research Assistant

Sydney, Australia

- Engage with clinicians to define research objectives and data acquisition protocols.
- Develop advanced deep learning algorithms for medical image analysis.
- Assist in writing research papers, reports, and other documents.
- Prepare materials for academic presentations, conferences, or seminars.

James Cook University

Visiting Research Student

From 10 2024
Cairns, Australia

- Conduct a comprehensive literature review on vision-language models (VLMs).
- Assist in writing research papers, summaries, and research proposals for funding applications.
- Help design and fine-tune vision-language models for specific medical imaging tasks (e.g., disease diagnosis).

COMPETITIONS

- **1st** Place Winner in the Open Development Phase and **2nd** Place Winner in the Closed Testing Phase of the PI-CAI Challenge **2022**
- **1st** Place Winner of Hitachi Video Analytics Technology Hackathon **2019**
- **2nd** Prize (Honorable Mention Award) in the Mathematical Contest in Modeling (MCM) **2018**
- **2nd** Prize in the Shaanxi Division of the National Undergraduate MCM **2017**
- **3rd** Prize in the 10th NPU New Concept Marine Vehicle Design Competition **2018**
- **3rd** Prize in the 17th NPU Undergraduate MCM **2016**
- **6th** Place Winner of the MICCAI LLD-MMRI2023 Challenge **2023**

HONORS & AWARDS

- Faculty of Engineering Research Scholarship (ERS), USYD **2020–2024**
- China National Scholarship, Ministry of Education of China (Top 0.2%) **2018**
- Scholarship of Outstanding Student (1st class), NPU **2016&2018**
- Advanced Individual Award in the May 4th Commendation, NPU **2016&2017&2018**
- Xianzi Zeng Education Foundation's 6th Outstanding University Student Award Program **2016&2017&2018**

LEADERSHIP & VOLUNTEER EXPERIENCES

- Reviewer for CMPB, ISBI'24; and helped to review for TMI, JBHI, MICCAI'23
- Secretary of IEEE USYD Student Branch **From 2023**
- Counselor Assistant & Student Affairs Assistant & Junior Class Assistant, NPU **2015–2019**
- Member of the 5th "Dream Pursuit Class" under the Soaring Talent Project, NPU **2017–2018**
- Founder and Chair of Student Literature Society, NPU **2016–2017**
- Outstanding Trainee, Career and Leadership Development Program, NPU **2016**
- Outstanding Volunteer in the 2nd China Space Day **2017**
- Outstanding Volunteer in the 6th National Marine Vehicle Design and Production Competition **2017**
- National-level Outstanding Practice Team in Social Practice—"The Love for Rongshui County, the Focus on Development" **2016**

PUBLICATIONS

Journal Articles

- [1] **Y. Yuan**, E. Ahn, D. Feng, M. Khadra, and J. Kim, *Z-SSMNet: Zonal-aware Self-supervised Mesh Network for Prostate Cancer Detection and Diagnosis with Bi-parametric MRI*, *Computerized Medical Imaging and Graphics*, 2025. **(IF: 5.4)** [Link]
- [2] **Y. Yuan**, E. Ahn, and J. Kim, *PLN: Pyramid Lattice Network with Self-supervised Pre-training for Diagnosis of Focal Liver Lesions using Anisotropic Multiparametric MRI*, 2024. **(In Preparation)**
- [3] **Y. Yuan**, E. Ahn, M. Meng, C. Jiang, S. Song, and J. Kim, *MTL-JSP: Multi-Task Learning for Joint tumor Segmentation and recurrence Prediction of Pancreatic Ductal Adenocarcinoma Using Preoperative ^{18}F -FDG PET/CT Images*, 2024. **(In Preparation)**
- [4] C. Jiang*, **Y. Yuan***, B. Gu, E. Ahn, J. Kim, D. Feng, Q. Huang, and S. Song, *Preoperative Prediction of Microvascular Invasion and Perineural Invasion in Pancreatic Ductal Adenocarcinoma with ^{18}F -FDG PET/CT Radiomics Analysis*, *Clinical Radiology*, 78(9), 687-696, 2023. **(IF: 2.1, * means Equal Contribution)**[Link]
- [5] A. Saha, J. S. Bosma, ..., and **the PI-CAI Consortium**, *Artificial Intelligence and Radiologists in Prostate Cancer Detection on MRI (PI-CAI): An International, Paired, Non-inferiority, Confirmatory Study*, *The Lancet Oncology*, 25(7), 879-887, 2024. **(IF: 41.6, contributed as an author within the PI-CAI Consortium)** [Link]
- [6] J. J. Twilt, A. Saha, J. S. Bosma, D. Yakar, M. Elschot, J. Veltman, J. Fütterer, H. Huisman, M. de Rooij and **the PI-CAI Consortium**, *Autonomous AI-Assisted Rule-In/Rule-Out System to Optimize Diagnosis and Workflow Efficiency for Clinically Significant Prostate Cancer on MRI*, 2024. **(Under Review, contributed as an author within the PI-CAI Consortium)**
- [7] R. Mayer, **Y. Yuan**, J. Udupa, B. Turkbey, P. Choyke, D. Han, H. Lin, and C. B. Simone II, *Comparing and Combining Artificial Intelligence and Spectral/Statistical Approaches for Elevating Prostate Cancer Assessment in Bi-Parametric MRI: A Pilot Study*, *Diagnostics*, 2025. **(IF: 3.0, Under Review)** [Link]
- [8] X. Qiao, C. Jiang, P. Li, **Y. Yuan**, Q. Zeng, L. Bi, S. Song, J. Kim, D. Feng, and Q. Huang, *Improving Breast Tumor Segmentation in PET via Attentive Transformation Based Normalization*, *IEEE Journal of Biomedical and Health Informatics*, 26(7), 3261-3271, 2022. **(IF: 6.7)** [Link]

Conference Articles

- [1] **Y. Yuan**, E. Ahn, D. Feng, M. Khadra, and J. Kim, *SSPT-bpMRI: A Self-supervised Pre-training Scheme for Improving Prostate Cancer Detection and Diagnosis in Bi-parametric MRI*, in *45th Annual International Conference of the IEEE Engineering in Medicine Biology Society (EMBC)*. IEEE, 2023: 1-4. [Link]
- [2] **Y. Yuan**, B. Sheng, P. Li, L. Bi, J. Kim, and E. Wu, *Deep Intrinsic Image Decomposition Using Joint Parallel Learning*, in *Proceedings of the 36th Computer Graphics International Conference*, CGI 2019: 336-341. [Link]
- [3] Y. Xia, **Y. Yuan**, E. Ahn, and J. Kim, *Multi-Phase and Hierarchical Unsupervised Learning Framework for Glioblastoma Sub-Region Segmentation in MRI Sequences*, in *25th International Conference on Digital Image Computing: Techniques and Applications (DICTA 2024)*. **(Oral)** [Link]